

MTH202 Final term Quiz 3

By Vusqa's EduQuest

MTH202 - Discrete Mathematics (Quiz No.3)

Question # 1 of 10 (Start time: 12:02:45 PM, 03 January 2025)

To prove by mathematical induction for $1+5+9+\dots+(4n-3)=n(2n-1)$ for all positive integers, the inductive step is true for $n=k$ is _____.

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☐ | $1+5+9+\dots+(4k+3)=k(2k-1)$ |
| ☐ | $1+5+9+\dots+(4k-3)=k(2k+1)$ |
| ☐ | $1+5+9+\dots+(4k+3)=k(2k+1)$ |
| ☒ | $1+5+9+\dots+(4k-3)=k(2k-1)$ |

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Question # 2 of 10 (Start time: 12:03:20 PM, 03 January 2025)

If 'n' is an odd integer then n^3+n is

Select the correct option

☐	odd
☒	even

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Question # 3 of 10 (**Start time: 12:03:50 PM, 03 January 2025**)

If the square of an integer is even , then that integer is

Select the correct option

☐	either even or odd
☒	even
☐	neither even nor odd
☐	odd

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Question # 4 of 10 (Start time: 12:04:18 PM, 03 January 2025)

The sum of $(6 - 7\sqrt{2})$ and $(6 + 7\sqrt{2})$ is

Select the correct option

☒	12
☐	$2(6-7\sqrt{2})$
☐	$12-14\sqrt{2}$
☐	$2(6+7\sqrt{2})$

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Question # 5 of 10 (Start time: 12:04:46 PM, 03 January 2025)

Which of the following statement is true according to the Division Algorithm?

Select the correct option

☐	$17 = 5 \times 4 - 3$
☐	$17 = 5 \times 5 - 8$
☒	$17 = 5 \times 3 + 2$
☐	$17 = 5 \times 1 + 12$

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Question # 6 of 10 (Start time: 12:05:17 PM, 03 January 2025)

The rational number '2' is the product of which of the two irrational numbers.

Select the correct option

☒	$\sqrt{2}, \sqrt{2}$
☐	$\sqrt{2}, \sqrt{3}$
☐	$\sqrt{2}, \sqrt{7}$
☐	$\sqrt{2}, \sqrt{5}$

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Question # 7 of 10 (Start time: 12:05:43 PM, 03 January 2025)

To prove by mathematical induction basis step is _____.

Select the correct option

☐	The proposition $P(1)$ is false
☒	The proposition $P(1)$ is true

Question # 8 of 10 (Start time: 12:06:10 PM, 03 January 2025)

To prove by mathematical induction for $1+5+9+\dots+(4n-3)=n(2n-1)$ for all positive integers, the basis step is _____.

Select the correct option

☒	$1=1(2(1)-1)$
☐	$1=1(2(1)+1)$

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Question # 9 of 10 (**Start time: 12:06:41 PM, 03 January 2025**)

The predicate describing the final state is called _____.

Select the correct option

☐	None
☐	Input variables
☒	post-condition of the algorithm
☐	pre-condition of the algorithm



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Question # 10 of 10 (Start time: 12:07:10 PM, 03 January 2025)

The sum of two odd integers is

Select the correct option

☐	odd
☒	even

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